B. Tech. (Computer Engineering) Vth Semester

2nd Sessional 2022-23, AUTOMATA THEORY (CEN-502)

Time: 1 Hour MM:15

Note: Attempt all questions. Date: 02/11/22

Your Roll No : .....

Q No.	CO Mapped
1	3
2	3
3	4

Q. No	Question	Marks
1.	*Using suitable reasons*, comment of the Ambiguity of the following CFGs a. $S \rightarrow \varepsilon \mid aS \mid aSbS$ b. $S \rightarrow aB \mid bC \quad A \rightarrow a \quad B \rightarrow b \quad C \rightarrow aBC \mid bAS$	4
2.	Convert the following CFG to CNF along with mentioning appropriate rules wherever needed $E \rightarrow EY \mid XeYe$ $X \rightarrow tYE \mid t \mid \varepsilon$ $Y \rightarrow Ey \mid XY \mid \varepsilon$	7
3	Construct a Deterministic PDA for the Language with all odd-length strings over $\{a,b\}$ with middle symbol a. ***Verify your answer with suitable examples.***	4